Mock diagnostic examination - Probability and Statistics full solutions

Full solutions

Use these solutions only after completing the 90-minute attempt. Review each method, result and common mistake, then record the first cause of any lost mark.

Section S0 — short diagnostic questions

Question 1

Part A — `MOCK-S0-Q01-PA` (S1) Method and conditions. Choose one person from each occupational group. The choices are independent stages of the counting procedure.

$$\binom{5}{1}\binom{3}{1} = 5 \times 3 = 15.$$

Final result. There are $\boxed{15}$ committees.

Marks (2). Correct multiplication-principle setup: 1. Correct count: 1.

Common mistakes. Choosing two people from all eight without enforcing the composition; dividing by 2! even though the roles come from distinct groups; adding $5 + 3$ rather than multiplying.

Question 2

Part A — `MOCK-S0-Q02-PA` (S3) Method and conditions. Under the course parameterisation, the exponential parameter is the rate and the support is non-negative.

The support is $x \in [0, \infty)$. Therefore

$$\mathbb{P}(X > 1) = \int_1^\infty 2e^{-2x}\,\mathrm{d}x = e^{-2}.$$

Final result. The support is $[0, \infty)$, and $\boxed{\mathbb{P}(X > 1) = e^{-2}}$.

Marks (2). Correct support: 1. Correct exact survival probability: 1.

Common mistakes. Treating 2 as the mean rather than the rate; giving the support as all real numbers; using $1 - e^{-2}$, which is $\mathbb{P}(X \leq 1)$.

Question 3

Part A — `MOCK-S0-Q03-PA` (S4, S6) Method and conditions. Use linearity of expectation and the independence of the identically distributed observations when calculating the variance.

$$\mathbb{E}[\bar{X}] = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{1}{16^2}\sum_{i=1}^{16}\mathrm{Var}(X_i) = \frac{4}{16} = \frac{1}{4}.$$

Final result.

$$\boxed{\mathbb{E}[\bar{X}] = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{1}{4}}.$$

Marks (2). Correct expectation: 1. Correct variance with the $1/n$ factor: 1.

Common mistakes. Dividing the variance by n^2 without summing all n variances; reporting the variance of an individual observation; confusing variance $1/4$ with standard deviation $1/4$.

Question 4

Part A — `MOCK-S0-Q04-PA` (S7) Method and conditions. Distinguish the fixed population parameter, the random estimator before data are observed, and the numerical estimate after observation.

The parameter is the fixed but unknown Poisson rate $\lambda > 0$. The estimator is

$$\hat{\lambda} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

For the observed sample,

$$\hat{\lambda}(\mathbf{x}) = \frac{1+0+2+1}{4} = 1.$$

Final result. Parameter: λ. Estimator: $\hat{\lambda} = \bar{X}$. Estimate: $\boxed{1}$.

Marks (2). Correct distinction between parameter and estimator: 1. Correct numerical estimate: 1.

Common mistakes. Calling the observed value 1 the estimator; treating λ as random; dividing the sample total by the wrong sample size.

Question 5

Part A — `MOCK-S0-Q05-PA` (S9) Method and conditions. Compare the p-value with the stated significance level and interpret only the evidence supplied by that test.

Because $0.08 > 0.05$, we fail to reject H_0 at the 5% level. This does not prove that H_0 is true. A false null can fail to be rejected when the sample does not provide sufficiently strong evidence against it; this is a Type II error.

Final result. $\boxed{\text{Fail to reject } H_0}$; the result is not proof that H_0 is true.

Marks (2). Correct decision: 1. Correct evidential explanation: 1.

Common mistakes. Rejecting because the p-value is numerically positive; writing "accept H_0" as proof; interpreting the p-value as the probability that H_0 is true.

Section S1 — probability, random variables and joint distributions

Question 6 — total probability and Bayes' rule

Part A — `MOCK-S1-Q06-PA` (S1, S2) Method and conditions. Partition the population into A and A^c, then use Bayes' rule with the total probability as the denominator.

$$\begin{aligned}\mathbb{P}(D) &= \mathbb{P}(D \mid A)\mathbb{P}(A) + \mathbb{P}(D \mid A^c)\mathbb{P}(A^c)\\ &= \frac{3}{4}\frac{3}{5} + \frac{1}{2}\frac{2}{5} = \frac{9}{20} + \frac{4}{20} = \frac{13}{20}.\end{aligned}$$

Bayes' rule gives

$$\mathbb{P}(A \mid D) = \frac{\mathbb{P}(D \mid A)\mathbb{P}(A)}{\mathbb{P}(D)} = \frac{9/20}{13/20} = \frac{9}{13}.$$

Final result.

$$\boxed{\mathbb{P}(D) = \frac{13}{20}, \qquad \mathbb{P}(A \mid D) = \frac{9}{13}}.$$

Among students who complete the task, 9/13 followed route A in this probability model.

Marks (5). Correct total-probability setup: 1. Correct $\mathbb{P}(D)$: 1. Correct Bayes numerator and denominator: 1. Correct $\mathbb{P}(A \mid D)$: 1. Correct conditional interpretation: 1.

Common mistakes. Using 3/5 as $\mathbb{P}(A^c)$; omitting the A^c branch; reversing $\mathbb{P}(A \mid D)$ and $\mathbb{P}(D \mid A)$; interpreting 9/13 among all students rather than completers.

Question 7 — a joint probability mass function

Part A — `MOCK-S1-Q07-PA` (S3, S4) Method and conditions. Check non-negativity and total mass, sum across rows or columns for the marginals, and then use the marginal PMFs for the expectations.

All entries are non-negative and

$$\frac{2+1+1+1+3+4}{12} = 1,$$

so the table is a valid joint PMF. The marginal PMFs are

$$p_X(0) = \frac{4}{12} = \frac{1}{3}, \qquad p_X(1) = \frac{8}{12} = \frac{2}{3},$$

and

$$p_Y(0) = \frac{3}{12} = \frac{1}{4}, \qquad p_Y(1) = \frac{4}{12} = \frac{1}{3}, \qquad p_Y(2) = \frac{5}{12}.$$

Therefore

$$\mathbb{E}[X] = 0\left(\frac{1}{3}\right) + 1\left(\frac{2}{3}\right) = \frac{2}{3},$$

and

$$\mathbb{E}[Y] = 0\left(\frac{1}{4}\right) + 1\left(\frac{1}{3}\right) + 2\left(\frac{5}{12}\right) = \frac{7}{6}.$$

Final result. The table is valid, with the marginals displayed above, and

$$\boxed{\mathbb{E}[X] = \frac{2}{3}, \qquad \mathbb{E}[Y] = \frac{7}{6}}.$$

Marks (5). Validity check: 1. Correct marginal PMF of X: 1. Correct marginal PMF of Y: 1. Correct $\mathbb{E}[X]$: 1. Correct $\mathbb{E}[Y]$: 1.

Common mistakes. Checking only that the entries are non-negative; summing the wrong row or column; omitting zero-valued support points from a PMF without specifying the support; calculating expectations from probabilities that do not sum to one.

Part B — `MOCK-S1-Q07-PB` (S5) Method and conditions. Divide the relevant joint masses by $p_X(1)$, retain the conditional support, and test independence with any valid numerical comparison, such as joint versus product or conditional versus marginal probability.

Because $p_X(1) = 8/12$, the conditional PMF is

$$p_{Y|X}(y \mid 1) = \begin{cases} 1/8, & y = 0, \\ 3/8, & y = 1, \\ 1/2, & y = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Hence

$$\mathbb{E}[Y \mid X = 1] = 0\left(\frac{1}{8}\right) + 1\left(\frac{3}{8}\right) + 2\left(\frac{1}{2}\right) = \frac{11}{8}.$$

The variables are not independent. For example,

$$\mathbb{P}(X = 1, Y = 0) = \frac{1}{12} \neq \mathbb{P}(X = 1)\mathbb{P}(Y = 0) = \frac{2}{3}\frac{1}{4} = \frac{1}{6}.$$

Equivalently, $\mathbb{P}(Y = 0 \mid X = 1) = 1/8 \neq 1/4 = \mathbb{P}(Y = 0)$.

Final result. The conditional PMF is displayed above, $\boxed{\mathbb{E}[Y \mid X = 1] = 11/8}$, and X and Y are not independent.

Marks (5). Correct conditioning denominator and support: 1. Correct conditional masses: 1. Correct conditional expectation: 1. Valid numerical independence comparison: 1. Correct independence conclusion: 1.

Common mistakes. Dividing by $p_Y(y)$ rather than $p_X(1)$; failing to renormalise the $x = 1$ row; using a conditional PMF that does not sum to one; declaring dependence without a numerical justification.

Section S2 — sampling distributions, the LLN and the CLT

Question 8 — exact and approximate sample-mean calculations

Part A — `MOCK-S2-Q08-PA` (S6) Method and conditions. A mean of independent normal variables is exactly normal. Calculate its mean and variance before standardising.

$$\mathbb{E}[\bar{X}] = 12, \qquad \operatorname{Var}(\bar{X}) = \frac{9}{9} = 1,$$

so

$$\bar{X} \sim \mathcal{N}(12, 1).$$

Therefore

$$\mathbb{P}(\bar{X} < 13) = \Phi\left(\frac{13 - 12}{1}\right) = \Phi(1) \approx 0.8413.$$

The normal sampling distribution and the exact probability expression $\Phi(1)$ are exact because the observations are independent normal random variables. The decimal 0.8413 is the supplied rounded value.

Final result.

$$\boxed{\bar{X} \sim \mathcal{N}(12, 1), \qquad \mathbb{P}(\bar{X} < 13) \approx 0.8413}.$$

Marks (4). Correct exact sampling distribution: 1. Correct standardisation: 1. Correct probability: 1. Correct exactness explanation: 1.

Common mistakes. Keeping variance 9 for the sample mean; using variance rather than standard deviation in the standardisation; invoking the CLT even though exact normality is available.

Part B — `MOCK-S2-Q08-PB` (S6) Method and conditions. Use the iid, finite-mean and finite-variance conditions for a large-sample normal approximation, then state the LLN as a separate convergence result.

For $n = 100$,

$$\bar{W} \dot{\sim} \mathcal{N}\left(50, \frac{100}{100}\right) = \mathcal{N}(50, 1).$$

Thus, using the supplied rounded normal-table value,

$$\begin{aligned} \mathbb{P}(48 < \bar{W} < 52) &\approx \Phi(2) - \Phi(-2) \\ &\approx 0.9772 - (1 - 0.9772) \\ &= 0.9544. \end{aligned}$$

Using a full-precision normal CDF instead gives 0.9545 to four decimal places; both values reflect the same correct standardisation and differ only because the supplied table is rounded.

The approximation uses independence, identical distribution, finite mean and finite variance, together with a sufficiently large sample. The law of large numbers states that

$$\bar{W} \xrightarrow{p} 50 \quad \text{as } n \to \infty.$$

Final result.

$$\boxed{\mathbb{P}(48 < \bar{W} < 52) \approx 0.9544} \quad \text{using the supplied table,} \qquad \boxed{\bar{W} \xrightarrow{p} 50}.$$

Marks (4). Correct approximate sampling distribution: 1. Correct standardised bounds and probability: 1. Relevant CLT conditions: 1. Correct, distinct LLN statement: 1.

Common mistakes. Using variance 100 rather than 1 for $\bar{W}$; calculating a one-sided tail; presenting the CLT as an exact result; confusing the LLN's convergence in probability with convergence of $\bar{W}$ to a normal distribution.

Section S3 — estimation and inference

Question 9 — estimator properties and method of moments

Part A — `MOCK-S3-Q09-PA` (S7) Method and conditions. Equate the first population and sample moments, then compare the repeated-sampling biases, variances and concentration of the two estimators.

For a Poisson variable, $\mathbb{E}[X_i] = \lambda$. Equating this moment to $\bar{X}$ gives

$$\hat{\lambda}_{\text{MOM}} = \bar{X}.$$

For $T_1 = \bar{X}$,

$$\text{Bias}(T_1) = \mathbb{E}[\bar{X}] - \lambda = 0, \qquad \text{Var}(T_1) = \frac{\lambda}{n}.$$

For $T_2 = X_1$,

$$\text{Bias}(T_2) = \mathbb{E}[X_1] - \lambda = 0, \qquad \text{Var}(T_2) = \lambda.$$

Both estimators are unbiased, but only T_1 is consistent. Since $\lambda/n \to 0$, Chebyshev's inequality implies $\bar{X} \xrightarrow{p} \lambda$. By contrast, the distribution and positive variance of $T_2 = X_1$ do not concentrate as n increases, so T_2 is not consistent for $\lambda > 0$.

Final result.

$$\boxed{\hat{\lambda}_{\text{MOM}} = T_1 = \bar{X} \text{ is unbiased and consistent}},$$

whereas $T_2 = X_1$ is unbiased but not consistent.

Marks (5). Correct moment equation and estimator: 1. Correct bias of T_1: 0.5. Correct variance of T_1: 0.5. Correct bias of T_2: 0.5. Correct variance of T_2: 0.5. Correct consistency conclusion for T_1: 0.5. Valid concentration argument for T_1: 0.5. Correct non-consistency conclusion for T_2: 0.5. Valid lack-of-concentration argument for T_2: 0.5.

Common mistakes. Equating λ to the sample variance instead of the requested first moment; calling an unbiased estimator automatically consistent; dividing $\text{Var}(X_1)$ by n; asserting that T_2 improves as unused observations are added.

Question 10 — a confidence interval and a two-sided test

Part A — `MOCK-S3-Q10-PA` (S8, S9) Method and conditions. The exact one-sample Student's t procedure requires an iid random sample from a normal population. The population variance is finite and unknown and is estimated by S^2.

The test statistic is

$$t = \frac{52.2 - 50}{4/\sqrt{16}} = \frac{2.2}{1} = 2.20,$$

with 15 degrees of freedom. The supplied two-sided p-value is approximately 0.0439. Since $0.0439 < 0.05$, reject $H_0 : \mu = 50$ at the 5% level in favour of $H_1 : \mu \neq 50$.

Using the supplied rounded critical value, the corresponding 95% confidence interval is

$$\begin{aligned} 52.2 \pm t_{0.975,15}\frac{4}{\sqrt{16}} &\approx 52.2 \pm 2.131(1) \\ &= [50.0690,\ 54.3310]. \end{aligned}$$

Using a full-precision $t_{0.975,15}$ instead gives $[50.0686, 54.3314]$. Both endpoint pairs are valid four-decimal answers for the stated method; the small difference is due only to the supplied critical value being rounded.

The hypothesised value 50 is outside this interval, which agrees with the rejection by the corresponding two-sided 5% test. In repeated sampling, 95% of intervals constructed by this procedure contain the fixed population mean μ. It is not correct to assign 95% probability to the fixed μ lying in this particular realised interval.

Final result.

$$\boxed{t = 2.20, \quad p\text{-value} \approx 0.0439, \quad \text{reject } H_0},$$

and the 95% confidence interval is

$$\boxed{[50.0690,\ 54.3310]} \quad \text{using the supplied critical value.}$$

Marks (7). Correct exact-procedure conditions: 1. Correct statistic and degrees of freedom: 1. Correct p-value decision: 1. Correct confidence- interval setup: 1. Correct endpoints: 1. Correct test–interval duality explanation: 1. Valid repeated-sampling interpretation: 1.

Common mistakes. Using a normal critical value despite unknown variance; omitting the normal-population condition for exactness; halving the supplied two-sided p-value; saying there is a 95% probability that the fixed mean lies in this realised interval.